

Arabic Summarization with a Seq2Seq Attention Model

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1. Introduction

Automatic text summarization is a core NLP task that aims to produce a concise and meaningful representation of a longer document while preserving its essential information. Two main paradigms exist: extractive summarization, which selects important sentences directly from the source text, and abstractive summarization, which generates new text capturing the semantic essence of the source. This project focuses on abstractive summarization for Arabic text, which presents unique challenges including morphological richness, dialect diversity, and limited NLP resources compared to English.

1.1 Objectives

By the end of this project, the team aimed to:

1. Understand and implement Seq2Seq architectures with encoder and decoder networks.
2. Incorporate attention mechanisms (Bahdanau/Additive Attention) to improve abstraction quality.
3. Perform Arabic-specific text preprocessing including normalization, tokenization, and tashkeel removal.
4. Train and evaluate a deep learning model on large-scale Arabic summarization datasets.
5. Apply ROUGE evaluation metrics to assess summary quality objectively.

2. Datasets

The project combined five Arabic summarization datasets to maximize training data diversity and coverage of both Modern Standard Arabic (MSA) and Egyptian dialect text.

Dataset	Source	Approx. Size	Description
Kaggle Arabic	Kaggle	~7,000	Arabic news articles with summaries (Processed Text / summarizer columns)
Summset	CSV corpus	~22,000	Arabic text-summary pairs with text / summarizer columns
SumArabic 1.0	Mendeley / JSONL	~84,000 (train/val/test)	MSA news articles with headlines as summaries; three-way split provided
Egyptian Arabic	Hugging Face / Parquet	~varies	Focused on Egyptian dialect; text / summarized text columns
AraSum (AbsSum)	GitHub / TSV	~49,600	Tab-separated monolingual Arabic corpus with article and summary pairs

2.1 Data Loading and Merging

Each dataset was loaded using a dedicated loader function that standardized column names to text (article body) and target (summary). Loaders handled encoding issues using `encoding_errors='replace'` and format-specific quirks:

- SumArabic: loaded line-by-line from JSONL files, parsing text and headline fields.
- AraSum: loaded as TSV with quote-ignoring (`csv.QUOTE_NONE`) to handle malformed rows.
- Egyptian Arabic: loaded directly from Parquet format, which handles encoding natively.

After loading all five sources, the datasets were concatenated and deduplicated on the (article, summary) pair. The final merged dataset contained 68,984 unique article-summary pairs ready for training.

3. Text Preprocessing

Arabic text requires specialized preprocessing that differs significantly from English NLP pipelines. A `clean_arabic_text()` function was implemented to normalize all text before tokenization.

3.1 Preprocessing Pipeline

1. URL and HTML removal: stripped hyperlinks and HTML tags using regex.
2. Bidirectional Unicode control character removal: removed invisible formatting characters (U+200F, U+202A, etc.) common in web-scraped Arabic text.
3. Alef normalization: mapped all Alef variants (اَ اِ اِى اِو) to the base form (ا) to reduce vocabulary size.
4. Ta marbuta normalization: mapped ة to ه for consistency.
5. Ya normalization: mapped final Ya (ي) to standard Ya (ي).
6. Tashkeel (diacritic) removal: stripped harakat and other diacritical marks (Unicode range U+064B–U+065F, U+0670).
7. Punctuation removal: removed all ASCII punctuation characters.

After cleaning, rows where either the article or summary became shorter than 10 characters were discarded to ensure training quality.

3.2 Tokenization

Rather than training a vocabulary from scratch, the project used the pre-trained AraBERT tokenizer (aubmindlab/bert-base-arabertv2), which provides a vocabulary optimized for Arabic text using WordPiece subword tokenization. Key parameters:

- Maximum article length: 200 tokens (truncated/padded)
- Maximum summary length: 60 tokens (truncated/padded)
- Vocabulary size: ~64,000 tokens (inherited from AraBERT)

For training the decoder, target sequences were shifted: the decoder input is tokens 0 to N-2, and the target output is tokens 1 to N-1 (teacher forcing setup).

3.3 Data Splits

The preprocessed data was split using scikit-learn's `train_test_split` with a fixed random seed (42):

- Training set: 80% (~55,187 samples)
- Validation set: 10% (~6,899 samples)
- Test set: 10% (~6,898 samples)

4. Model Architecture

The system implements a Bidirectional Sequence-to-Sequence model with Additive (Bahdanau) Attention using TensorFlow and Keras. Pre-trained AraBERT embeddings are used as frozen initializers, significantly reducing trainable parameters while leveraging rich Arabic word representations.

4.1 Encoder

5. Input: token IDs of shape (batch, 200)
6. Embedding layer: frozen AraBERT embeddings (768-dim), with `mask_zero=True` to handle padding
7. Bidirectional LSTM: 128 units in each direction (256 total). Returns full sequences and final states.
8. State concatenation: forward and backward hidden states (h , c) are concatenated to form the initial decoder state.

4.2 Decoder

Input: shifted target token IDs of shape (batch, 59)

Embedding layer: same frozen AraBERT embeddings (shared with encoder)

LSTM: 256 units, initialized with concatenated encoder final states. Returns full output sequences.

4.3 Attention

Additive (Bahdanau) Attention is applied between decoder outputs and encoder outputs. The attention context vector is computed as a weighted sum of encoder outputs, where weights reflect how relevant each encoder time step is to the current decoder time step. The context vector is concatenated with the decoder LSTM output before the final dense layer.

4.4 Output

A TimeDistributed Dense layer with softmax activation over the full vocabulary produces a probability distribution per decoder time step. The layer explicitly uses float32 dtype for numerical stability under mixed-precision training.

5. Training

Hyperparameter	Value
Batch Size	16 (kept small for GPU memory safety)
Max Epochs	8
Optimizer	Adam with default learning rate
Early Stopping Patience	4 epochs on validation loss
LR Reduction Factor	0.5 (patience: 2 epochs, min lr: 1e-6)
Model Checkpoint	Best validation loss saved to best_seq2seq_model.keras
History Logging	CSV logger saved epoch metrics to training_history.csv

6. Evaluation

6.1 ROUGE Metrics

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) is the standard metric for text summarization. Three variants were computed:

- ROUGE-1: unigram overlap between generated and reference summary.
- ROUGE-2: bigram overlap, measuring fluency and coherence.
- ROUGE-L: longest common subsequence, capturing sentence-level structure.

The F-measure (harmonic mean of precision and recall) was used as the primary score. Evaluation was performed on a random sample of 20 test examples using the rouge_score library.

A second refined evaluation script additionally applied the Arabic text cleaning function to both generated and reference summaries before scoring, ensuring fair comparison by eliminating tokenization artefacts (such as ## WordPiece subword prefixes) from the comparison.

Our codes rouge evaluation gave us 0, due to the exact matching that rouge follows between reference and generated outputs. Our generated output could understand what the article was on, but could generate the words that exactly match the context.

[Sample 1]

Article: لحق رودا كيركراده بفيينورد روتردام الى المباراة النهائية لمسابقة كأس هولندا في كرة القدم بفوزه على مضيفه ... هيراكليس الميلو بركلات الترجيح 3-5 بعد انتهاء الوقتين الاصلي والاضافي بالتعادل 2-2

Reference: رودا إلى نهائي كأس هولندا

Generated: كأس الاتحاد في دوري ابطال اوربا

[Sample 2]

Article: اعلن نبيل بسباس رئيس الغرفة النقابية لتجار المواد الاوليه والاحذيه والجلود خلال اجتماع عام مشترك عقد ...اليوم الاربعاء لتدارس الوضعيه المترديه ال اليها قطاع الجلود والاحذيه تراجع الانتاجيه ارباعها وقال ب

Reference: كما يشكو القطاع من نقص في اليد العاملة وندرة في اليد العاملة المختصة .

Generated: ويأتي هذا الاجتماع القضائي الاجتماعي في قطاعات الدولة العموميه ، وفق بلاغ صادر عن وزاره الدفاع الوطني للاستثمار في التعليم الحالي

[Sample 3]

Article: اجلت الدائر الجناثيه بالمحكمه الابتدائيه بقرمباليه اليوم النظر قضيه حرق منزل شقيقه الرئيس الاسبق ...نعيه بن علي بحمام بنت الجديدي الى جلسه يوم افريل القادم استجابه لطلب دفاع المتهمين يذكر شخصا صادره حق

Reference: يذكر أن 18 شخصا صادرة في حقهم أحكام غيابية تراوحت بين عامين و 16 عاما اعترضوا على هذه الاحكام .وهم يمثلون في القضية بحالة سراح

Generated: تمكنت السلطات التونسيه للمرطه بالقطب الجزائري صباح اليوم الثلاثاء 27 فيفري 2017 ، اليوم الاربعاء عطله تسفير الشباب التونسي اليوم الجمعه

6.2 Qualitative Evaluation

A sanity check was performed by generating summaries for three randomly sampled test articles. Each output was compared side-by-side:

- The article text (first 200 characters)
- The reference (gold) summary
- The model-generated summary

7. Libraries

- TensorFlow / Keras: model definition, training, and inference
- Scikit Learn: train/validation/test splitting
- rouge_score: official ROUGE metric calculation
- pandas / NumPy: data loading, manipulation, and array operations
- Transformers

9. Discussion

Frozen pre-trained embeddings: Using AraBERT embeddings without fine-tuning was a deliberate choice to reduce GPU memory and training time. The 768-dimensional embeddings provide rich Arabic morphological and semantic information that a smaller model trained from scratch cannot replicate.

Bidirectional encoder: A bidirectional LSTM captures context from both directions of the input, which is especially important for Arabic where morphological agreement spans long distances.

Additive Attention: Bahdanau (additive) attention was chosen over Luong (multiplicative) attention because it tends to perform better in lower-resource training scenarios and is natively supported in Keras via `AdditiveAttention`.

Dataset merging strategy: Combining five different datasets improves domain coverage and reduces overfitting to any single source's style. Deduplication ensures the model does not see the same pair multiple times.

10. Limitations and Challenges

The ROUGE evaluation sample size of 20 examples is small; a larger evaluation set would yield more statistically reliable scores.

The maximum article length of 200 tokens may truncate long news articles, causing information loss.

Inference uses greedy decoding with temperature; beam search would likely produce higher ROUGE scores but was not implemented.

The model does not fully handle dialect diversity; while the Egyptian Arabic dataset is included, training data is predominantly MSA.

Cells 3 and 4 are duplicates of the qualitative check, reflecting iterative development in the notebook.

11. Conclusion

This project successfully implemented an abstractive Arabic text summarization pipeline using a Bidirectional LSTM Seq2Seq model with Additive Attention and pre-trained AraBERT embeddings. The pipeline covers all required components: multi-dataset loading and merging, Arabic-specific text preprocessing, tokenization with a pre-trained Arabic vocabulary, model training with early stopping and learning rate scheduling, inference with temperature-scaled greedy decoding, and quantitative ROUGE evaluation.

The use of pre-trained AraBERT embeddings as frozen initializers was a key design choice that leveraged rich Arabic semantic representations without the computational cost of full BERT fine-tuning. The combined dataset of nearly 69,000 article-summary pairs provides substantial coverage of both Modern Standard Arabic and Egyptian dialect text.

Future work should prioritize beam search decoding, gradual embedding unfreezing, and deploying the trained model in an interactive web interface to fulfill the Streamlit/Gradio deliverable requirement.